Agent Reply Latency – What the Numbers Mean and How to Reach 2.5 s

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In test call 58ca4b5f the caller waited a median **5.2 s** after speaking before the agent answered, and 1 turn in 10 waited **12.4 s** or more. Our target is **2.5 s**. Across all 12 test calls (117 turns) only **4 turns (3%)** met it. The AI model causes 77% of the wait, and 65% of turns run the model twice before the agent says anything, so the fix is in the agent design, not in the network or the voice.

The two numbers

The latency report shows two numbers for every turn. **Real wait** is the one we hold the agent to; **sum of parts** explains where that wait came from.

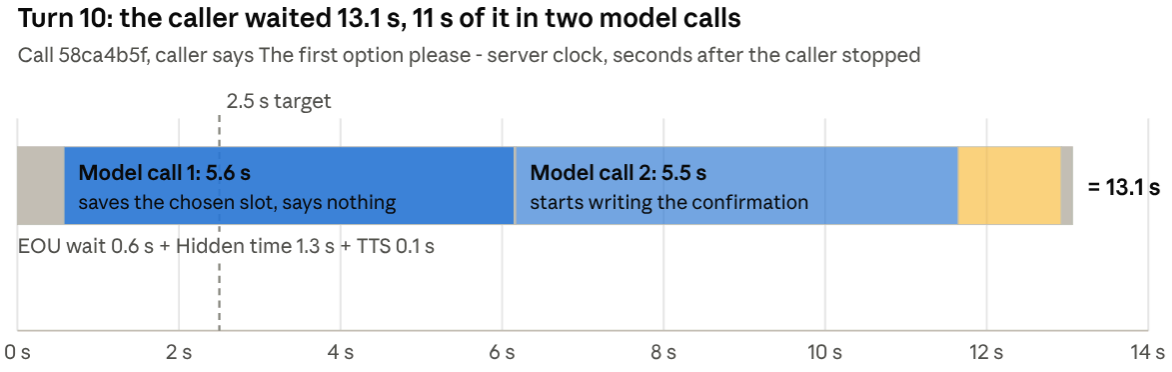
Call 58ca4b5f, 12 turns	P90	Median	Average	Fastest
Real wait (server clock)	**12.4 s**	**5.2 s**	**6.3 s**	**2.2 s**
Sum of parts (EOU + model + TTS)	11.5 s	4.6 s	5.5 s	1.6 s

- **Real wait** = a stopwatch on the server: the moment the caller stopped talking to the moment the agent's first audio was ready. Nothing is estimated; it is two timestamps subtracted.
- **Sum of parts** = the three timings the agent logs, added up: the end-of-turn wait (EOU), the time each model call takes to *start* writing (time to first token), and the time the voice engine takes to produce its first audio (TTS).

Real wait is always a little higher (here 0.6 s at the median) because the model timing stops at the first word. The caller also waits while the model finishes writing a step that only saves data, and while the reply reaches the voice engine. That leftover time is called **hidden time** below.

One real turn, step by step

The caller picked the first slot offered. The agent then ran the model twice before saying anything, and the caller heard silence for 13.1 s.



call 58ca4b5f turn 10, from flow_trace timestamps · blue = model, amber = hidden time, grey = EOU and TTS

Model call 1 (4,942 prompt tokens) only saved the slot. Model call 2 (2,691 tokens) began the confirmation, and 1.3 s more passed while the first sentence was written and handed to the voice. The sum of parts for this turn is 12.0 s; the real wait is 13.1 s.

Where the hidden time comes from

After the model writes its first word, the agent waits until a whole first sentence is written before it hands anything to the voice engine. No metric records that wait, but the caller hears it as silence. Across 132 turns it nets to about 0.5 s per turn.

Effect	What happens	Effect on the wait	Today
A. Waiting for a full first sentence	Text goes to the voice engine only once the first sentence is complete	Adds silence	Median 0.79 s, P90 1.34 s
B. Model starts early	The model starts before the end-of-turn check (EOU) finishes	Saves a little	Median 0.36 s, in 92% of turns

Real example, call `7e8f6c47` (19:39 IST), the caller says "I'd like to book a cardiology appointment, please.":

Time (UTC)	Event
14:09:25.76	Caller stops speaking
14:09:26.34	Agent confirms the caller is done (EOU 578 ms)
14:09:27.19	Model call 1 done: moves to the ask-identity step (1,349 ms)

Time (UTC)	Event
14:09:30.38	Model call 2 writes its first word (3,130 ms)
14:09:31.56	First sentence finished and handed to the voice: **1.18 s of hidden time**
14:09:31.67	First audio (TTS 109 ms); the agent starts speaking

Sum of parts is 5.2 s; the real wait is 5.9 s. To recover most of the hidden time (about 0.5-0.8 s a turn): send text to the voice engine at the first comma or short phrase instead of a full sentence, start replies with a very short sentence ("Sure.", "Got it, Anna."), and keep the early model start.

Where the time goes across all calls

Across 12 test calls and 117 caller turns, real wait was a median **4.7 s** and P90 **9.6 s**, and the model took most of it.

Part of the real wait	Median per turn	Share of all waiting
Model time to first token (all calls in the turn)	3.4 s	77%
Hidden time (model finishing, hand-over to voice)	0.5 s	10%
EOU wait (is the caller done?)	0.58 s	11%
TTS first audio	0.12 s	2%

Four patterns drive the model time:

- **Two model calls per turn.** 76 of 117 turns ran the model twice before speaking; the first call usually only moves to the next step or saves a value. Those turns wait a median **5.4 s**, against **3.2 s** with one call.
- **Large prompts.** Model start time grows with prompt size: under 3,000 tokens a median 1.8 s, 3,000-4,500 tokens 3.1 s, over 4,500 tokens 5.6 s. The slot steps (`collect_appointment_details`, `offer_alternatives`) send the largest prompts.
- **No prompt caching.** None of 249 model calls used cached tokens, so every call re-reads the whole prompt from scratch.
- **Slow steps stack.** Saving the chosen slot then confirming it is two full model calls in a row, which is how turn 10 above reached 13 s.

How to fix it

The changes are all in the agent, in this order of impact. Expected effects come from today's data where we have it; the rest must be measured after the change.

#	Change (agent dev team)	Evidence today	Expected effect
1	**One model call per turn.** Make step transitions rule-based, or let the same call save the value, move on and write the reply.	65% of turns call the model twice; 5.4 s vs 3.2 s median	About **2 s** off two-thirds of turns
2	**Turn on prompt caching** for the fixed part of each prompt (instructions, tools).	0 of 249 calls used the cache	Faster model start on every call; size to be measured
3	**Shrink prompts in the slot steps**: pass only the slots being offered, not all 12; trim instructions.	Under 3,000 tokens starts in 1.8 s; over 4,500 in 5.6 s	2-4 s off slot-search and booking turns
4	**Fixed replies for simple confirmations** ("Anna Beckett - is that right?") with no model call.	These turns still take 3-4 s	Those turns drop to about 1 s
5	**A faster model for simple steps** (greeting, identity, confirmations).	Simple steps still start in 1.1-2.7 s	To be measured per model
6	**Speak a short holding line at once** when a tool must run ("One moment, checking that").	Agent already does this, but only after a model call	Caller hears something within 1 s; does not shorten the work
7	**Send text to the voice in smaller pieces** (first comma or short phrase) and start replies with a very short sentence; tune EOU (577 ms) only after 1-5.	0.79 s median spent waiting for a full first sentence; EOU is 11% of the wait	0.5-0.8 s per turn from the voice change; at most 0.2-0.3 s from EOU

How to reach and prove 2.5 s

2.5 s is reachable only if each turn makes **one** model call that starts writing in about 1.6 s; today's one-call turns already get to 3.2 s.

Part	Today (median)	Budget for 2.5 s
EOU wait	0.58 s	0.5 s
Model, time to first token	3.4 s (often 2 calls)	1.6 s (1 call)
Hidden time	0.5 s	0.3 s
TTS first audio	0.12 s	0.1 s
Real wait	**4.7 s**	**2.5 s**

The path, measured after each step with the same test:

1. **Fix 1 (one call per turn)** - expected median about **3.2 s**, today's one-call figure.
2. **Fixes 3 and 4 (smaller prompts, fixed confirmations)** - prompts under 3,000 tokens start in 1.8 s, which puts the median near **2.7 s**.
3. **Fixes 2 and 5 (caching, faster model for simple steps)** - the last 0.2-0.5 s to reach **2.5 s**.

Pass criteria (proposed): over at least 30 test calls on the same agent, real wait median **⩽ 2.5 s**, P90 **⩽ 4 s**, and every booking confirmed. QA runs the test and shares the PDF after each agent release:

```
.\.venv\Scripts\python.exe webcall_load_test.py --livekit-url
ws://172.17.20.214:7880 --dynamic --patient-count 10 --hold-timeout 30
python call_latency.py --file results\dynamic_patient_*_YYYYMMDD_*.json --pdf
latency_after_fix.pdf
```

The **Real wait (server clock)** row in that PDF is the number to compare with 2.5 s.